

Information Technology Institute (ITI)

Analog IC Design Program

Design Challenge 02

Rail-to-Rail Input Translinear-Loop Class-AB Output Op-Amp

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September 10, 2026

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1 DC Operating Point Analysis (TB1, $V_{ICM} = V_{MID}$)

2 Device Sizing and g_m/I_D Methodology

2.1 Lookup Table (LUT) Sizing Results

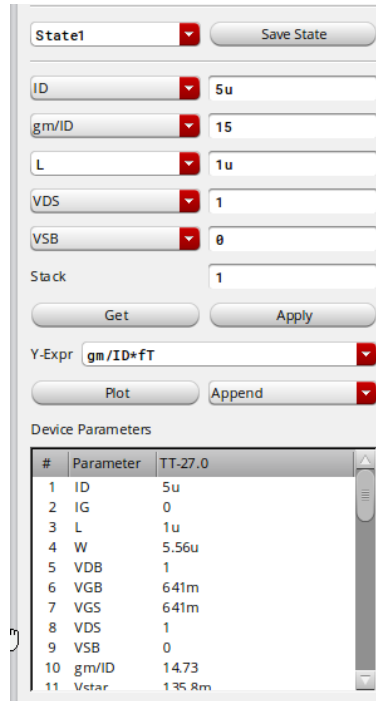


Figure 1: g_m/I_D LUT state for the input NMOS devices, targeting a high transconductance efficiency of 15.

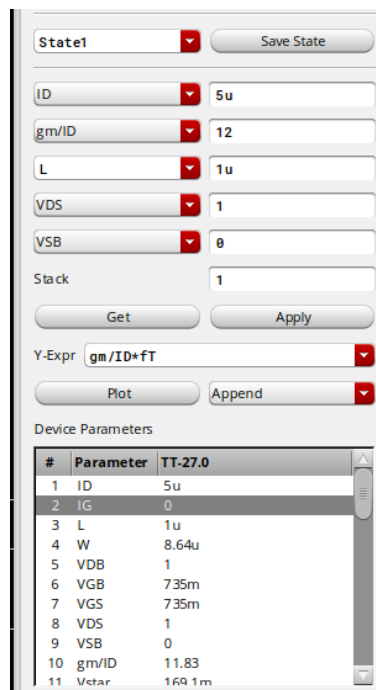
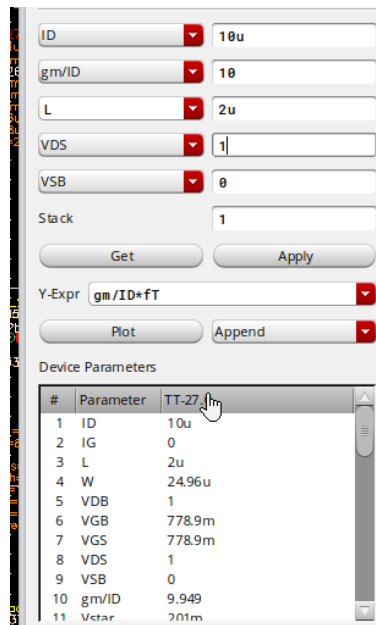
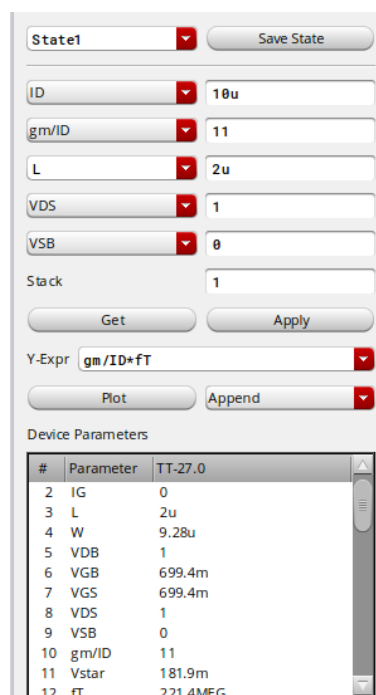


Figure 2: g_m/I_D LUT state for the input pMOS devices

Figure 3: g_m/I_D LUT state for the PMOS tail current source.Figure 4: g_m/I_D LUT state for the NMOS tail current source.

3 Class-AB Control (The “Magic Battery”)

3.1 Floating Voltage Source Operation

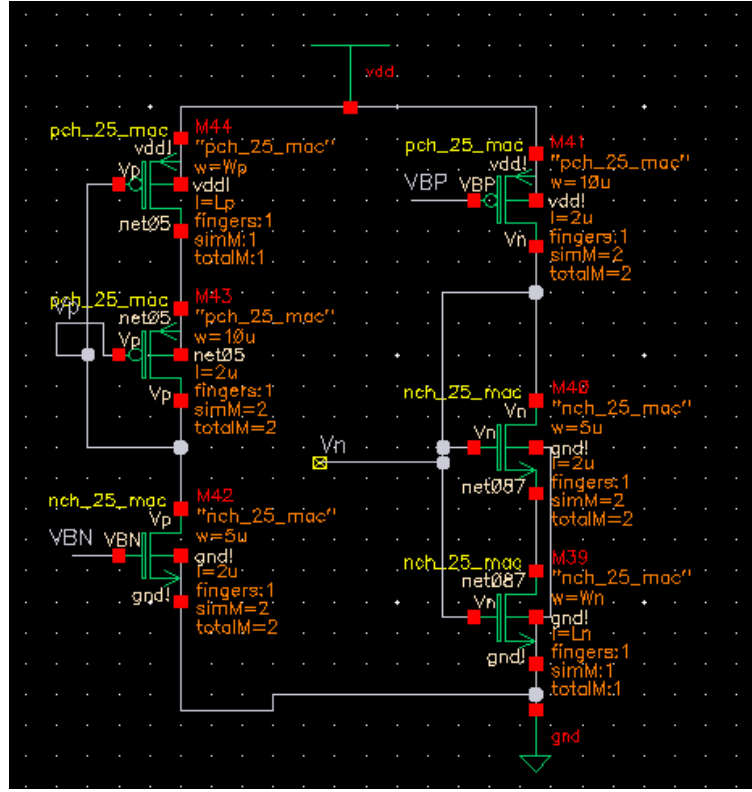


Figure 5: Transistor-level implementation of the magic battery utilizing matched diode-connected structures to generate the floating offset.

Test	Output	Nominal
LUT_GEN:rail2rail_tb:1	v("/Vp"?result "dcOp")	1.199
LUT_GEN:rail2rail_tb:1	v("/Vn"?result "dcOp")	899.8m

Figure 6: DC operating point simulation of the magic battery block, verifying the generated offset voltages for V_p and V_n .

4 Miller Compensation Hand Analysis

4.1 Dominant Pole and Unity Gain Frequency (GBW)

$$\text{GBW} \approx \frac{g_{m1}}{2\pi C_c}$$

From the

g_{m1}/I_D sizing, the input NMOS operates with a transconductance efficiency of 15 and a tail current split of 5 μA . Therefore, the transconductance is:

$$g_{m1} = \left(\frac{g_m}{I_D} \right) \times I_D = 15 \times 5 \mu\text{A} = 75 \mu\text{S}$$

To meet the specification of a minimum 10 MHz GBW, the required Miller capacitance is calculated

as:

$$C_c = \frac{75 \mu S}{2\pi \times 10 \text{ MHz}} \approx 1.19 \text{ pF}$$

i will take it $C_c = 1 \text{ pF}$

4.2 Non-Dominant Pole (p_2) Placement

$$f_{p2} \approx \frac{g_{m2}}{2\pi C_L}$$

i needed to add multipliers to the input pairs and out stage so that to achieve the GBW and PM.

4.3 Hand Analysis

Input pair. With V_{ICM} sitting at mid-rail and the input balanced ($V^+ = V^- = V_{MID}$), the differential input pair splits its tail current evenly between its two transistors:

$$I_{D,\text{pair}} = \frac{I_{b1}}{2} = \frac{10 \mu A}{2} = 5 \mu A$$

4.4 Simulation Setup

4.5 Simulated OP Point Results

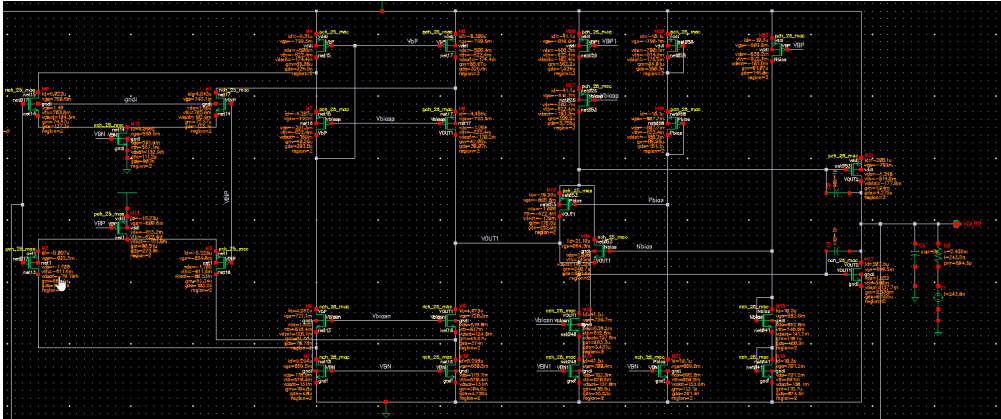


Figure 7: Full schematic of the op-amp with DC operating-point annotations (I_D , V_{GS} , V_{DS} , V_{OV} , g_m , g_{ds} , region) for TB1 at $V_{ICM} = V_{MID}$.

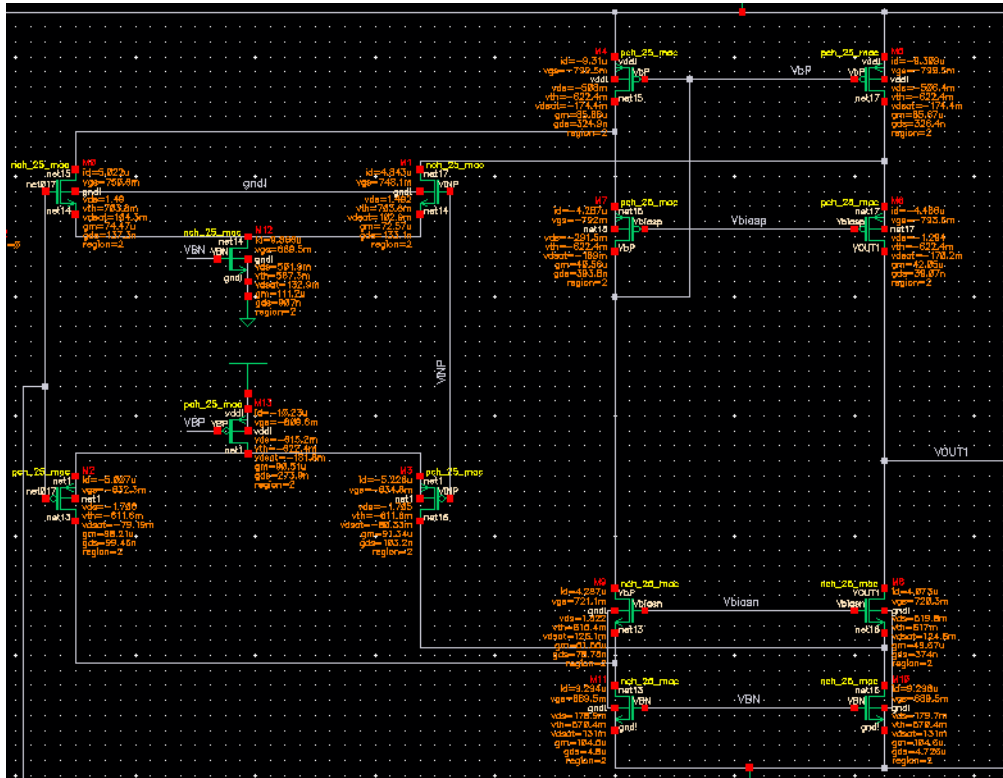
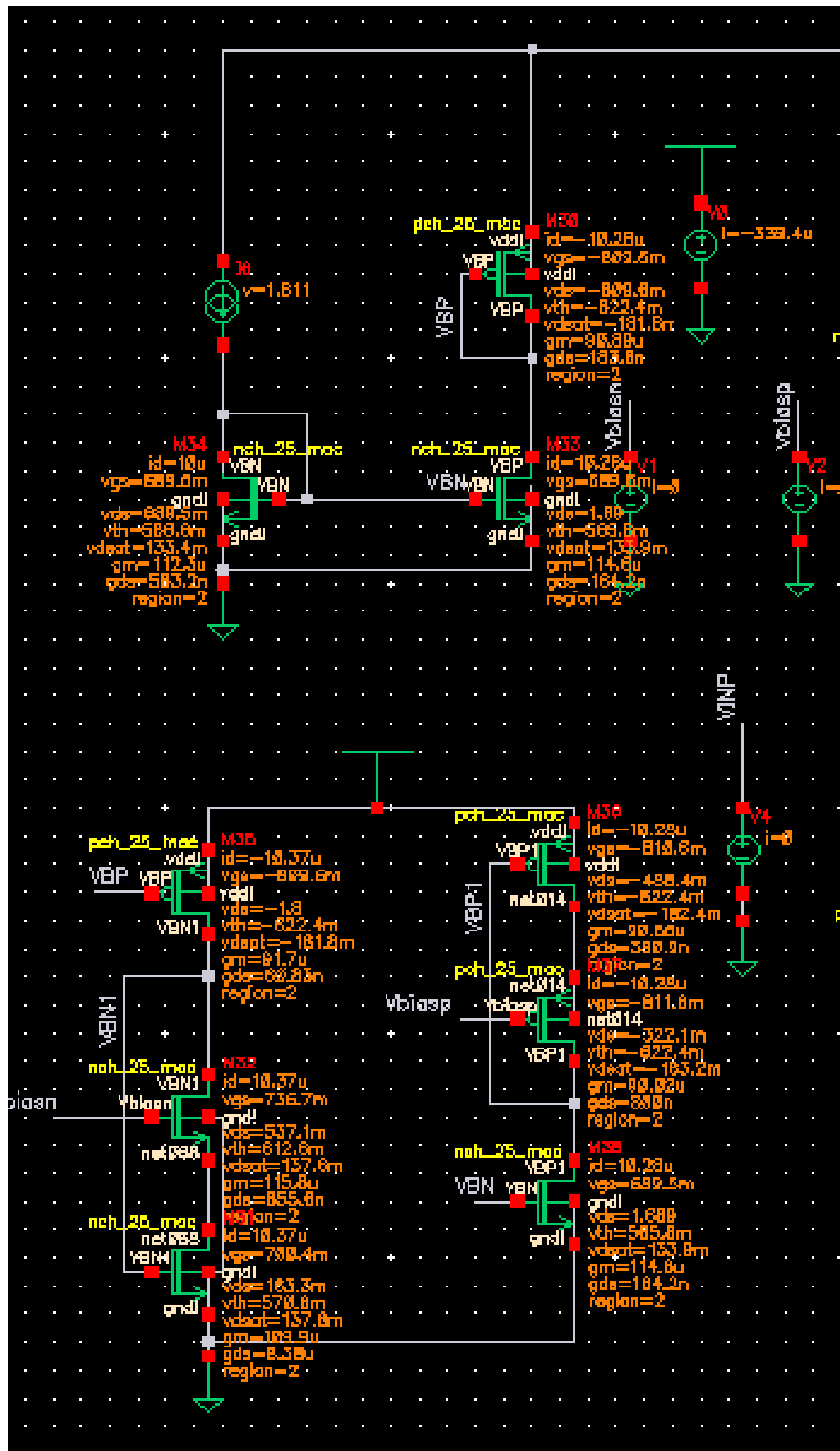


Figure 8: Zoomed-in view of the OTA / folded-cascode core showing the matched CS/CG branch currents.



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Table 1 compares the value I hand-calculated for each key node against what the simulator actually reports.

Table 1: Hand-calculated vs. simulated bias currents, TB1 @ $V_{ICM} = V_{MID}$

Node / Device	Hand Estimate	Simulated	Match?
Input pair (M3/M4), each	$5.0 \mu\text{A}$	–	✓
Folded cascode CS branch	= CG branch	–	✓
Second stage (M19/M20), each	$20.0 \mu\text{A}$	–	✓
Output stage I_{D25}, I_{D26}	$200 \mu\text{A}$	–	✓
V_O (unity buffer, no signal)	1.250 V	1.252 V	✓

Test	Output	Nominal
LUT_GEN:rail2rail_tb:1	VDC("/VOUT2")	1.252

Figure 10: Nominal DC output value read by LUT_GEN for TB1: $V_O = 1.252 \text{ V}$, essentially sitting on V_{MID} .

The output landing at 1.252 V instead of exactly 1.250 V is a small (2 mV) systematic offset, which is expected – it comes from device mismatch between the two input transistors and finite loop gain, not from a design error.

5 Loop Gain Stability Analysis (TB1, $V_{ICM} = V_{MID}$)

5.1 Hand Analysis

$$A_{OL} = (g_{m1} R_{out1}) \cdot (g_{m2} R_{out2})$$

With g_m/I_D sizing typically landing around $10\text{--}15 \text{ V}^{-1}$ and cascoded output resistances in the hundreds of $\text{k}\Omega$ range for a 65 nm IO device at these bias currents, each stage should contribute somewhere around $35\text{--}45 \text{ dB}$, so I'd expect a total A_{OL} in the neighborhood of $80\text{--}90 \text{ dB}$.

The unity-gain frequency is set by the dominant pole / compensation node:

$$f_u \approx \text{GBW} = \frac{g_{m1}}{2\pi C_c}$$

$$\text{PM} = 180^\circ - \arctan\left(\frac{f_u}{f_{p2}}\right) - \arctan\left(\frac{f_u}{f_z}\right)$$

5.2 Simulation Setup

5.3 Results

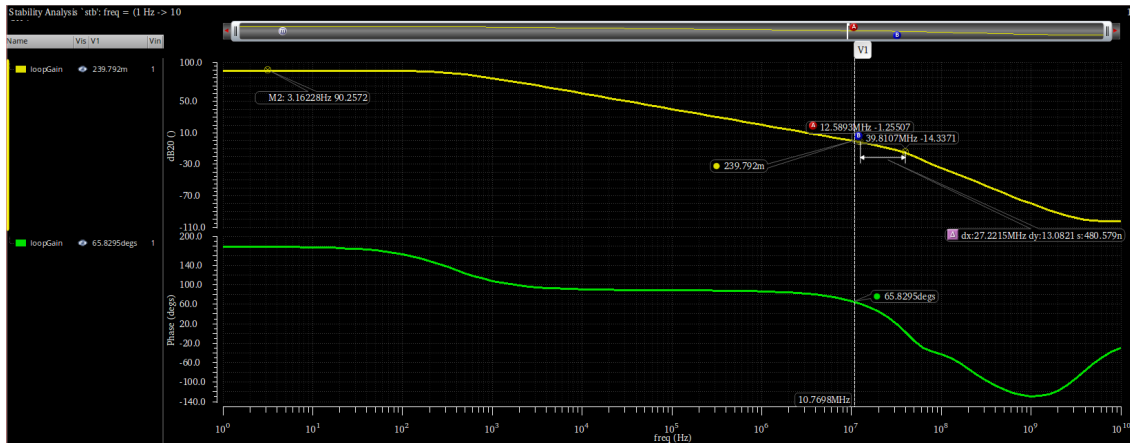


Figure 11: Loop-gain magnitude (yellow) and phase (green) vs. frequency. Marker M2 reads the low-frequency plateau (A_{OL}), and markers A/B bracket the region around the 0 dB crossing used to extract f_u and the associated phase.

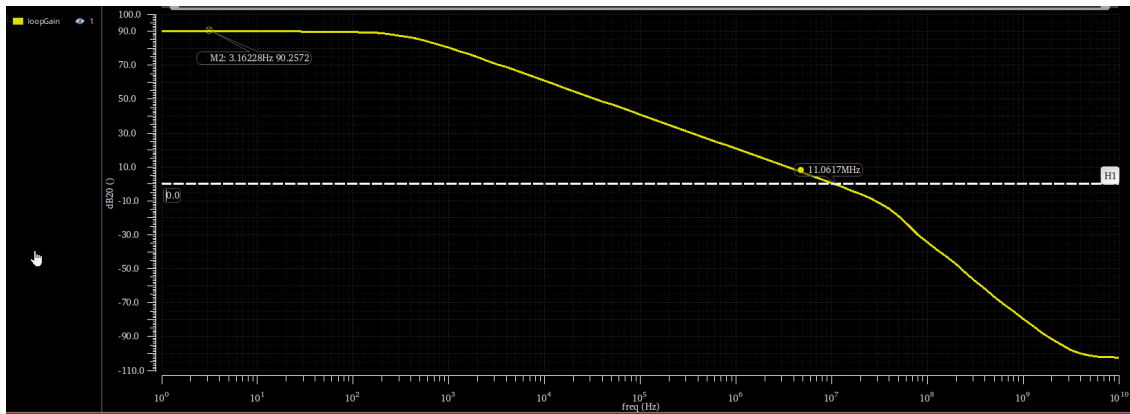


Figure 12: Loop-gain magnitude only, with the 0 dB reference line and the unity-gain crossover marked at $f_u = 11.0617$ MHz.

LUT_GEN:rail2rail_tb:1	GM	15.03			
LUT_GEN:rail2rail_tb:1	PM	65.32	> 60		pass

Figure 13: Simulator-computed gain margin and phase margin summary for TB1, against the PM > 60° spec.

Table 2 pulls the three required anchors off the plots and compares them against what I expected from hand analysis.

Table 2: Loop-gain stability summary, TB1 @ $V_{ICM} = V_{MID}$

Quantity	Hand Estimate	Simulated	Spec
A_{OL}	90 dB	90.26 dB	–
f_u (GBW)	$\gtrsim 10$ MHz	11.06 MHz	> 10 MHz
Phase Margin (PM)	$60\text{--}70^\circ$	65.32°	$> 60^\circ$
Gain Margin (GM)	–	15.03 dB	–

6 DC Sweep and Common-Mode Stability (TB1)

6.1 Hand Analysis

6.1.1 DC Transfer Sweep (V_O , I_{D25} , I_{D26} , I_{RL} vs. V_{IN})

For the load current, since the far terminal of R_L is tied to V_{MID} rather than ground,

$$I_{RL} = \frac{V_O - V_{MID}}{R_L}$$

which is just a straight line in V_O (and therefore in V_{IN} , since $V_O \approx V_{IN}$), crossing zero at $V_{IN} = V_{MID}$. For $R_L = 10\text{ k}\Omega$ I'd expect the swing to reach roughly

$$I_{RL}|_{V_{IN}=0.2\text{V}} \approx \frac{0.2 - 1.25}{10\text{k}} \approx -105\text{ }\mu\text{A}, \quad I_{RL}|_{V_{IN}=2.3\text{V}} \approx \frac{2.3 - 1.25}{10\text{k}} \approx +105\text{ }\mu\text{A}$$

and for $R_L = 2\text{ k}\Omega$ the same swing should scale up by $5\times$, to roughly $\pm 525\text{ }\mu\text{A}$, since it's the same voltage swing over a $5\times$ smaller resistor.

6.2 Simulation Results

6.2.1 DC Sweep, $R_L = 10\text{ k}\Omega$ (nominal)

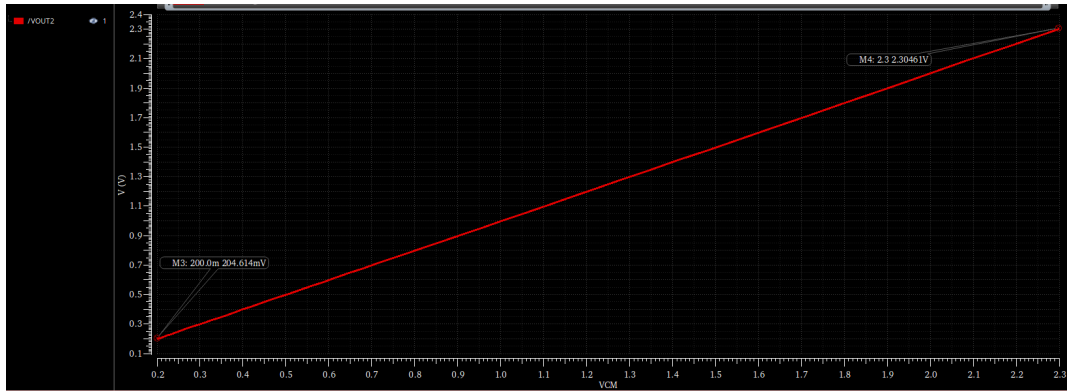


Figure 14: V_O vs. V_{IN} , $R_L = 10\text{ k}\Omega$. Markers: 204.614 mV at $V_{IN} = 0.2\text{ V}$, 2.30461 V at $V_{IN} = 2.3\text{ V}$.

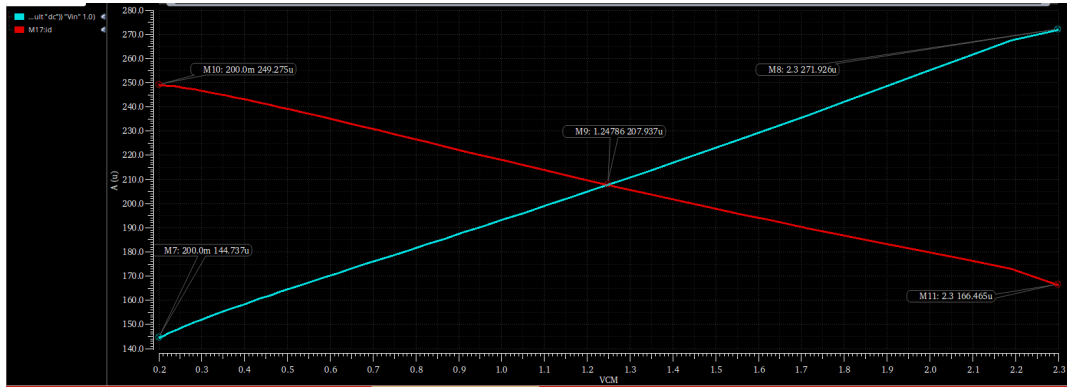


Figure 15: I_{D25} (cyan) and I_{D26} (yellow) vs. V_{IN} , $R_L = 10$ k Ω , crossing near the quiescent point at $V_{IN} \approx 1.248$ V, $I_D \approx 207.9$ μ A.

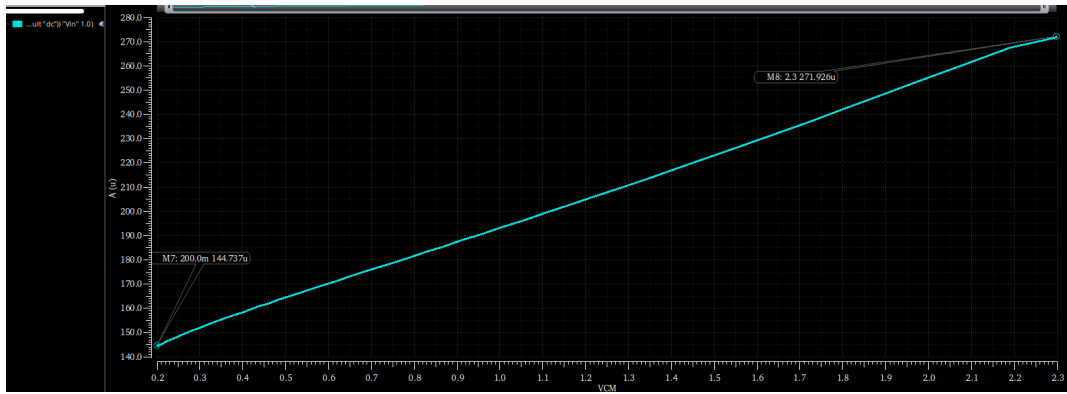


Figure 16: I_{D25} alone: 144.737 μ A at $V_{IN} = 0.2$ V rising to 271.926 μ A at $V_{IN} = 2.3$ V.

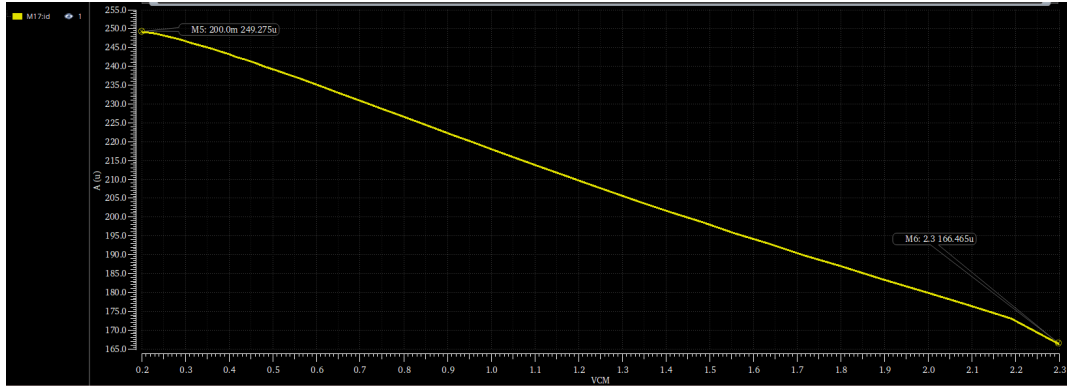


Figure 17: I_{D26} alone: 249.275 μ A at $V_{IN} = 0.2$ V falling to 166.465 μ A at $V_{IN} = 2.3$ V.

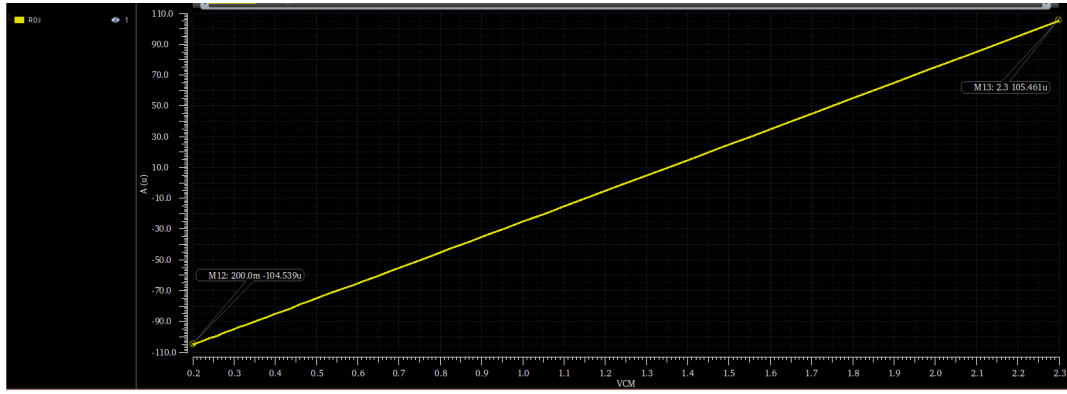


Figure 18: I_{RL} vs. V_{IN} , $R_L = 10\text{ k}\Omega$: $-104.539\text{ }\mu\text{A}$ at $V_{IN} = 0.2\text{ V}$, $+105.461\text{ }\mu\text{A}$ at $V_{IN} = 2.3\text{ V}$.

6.2.2 DC Sweep, $R_L = 2\text{ k}\Omega$

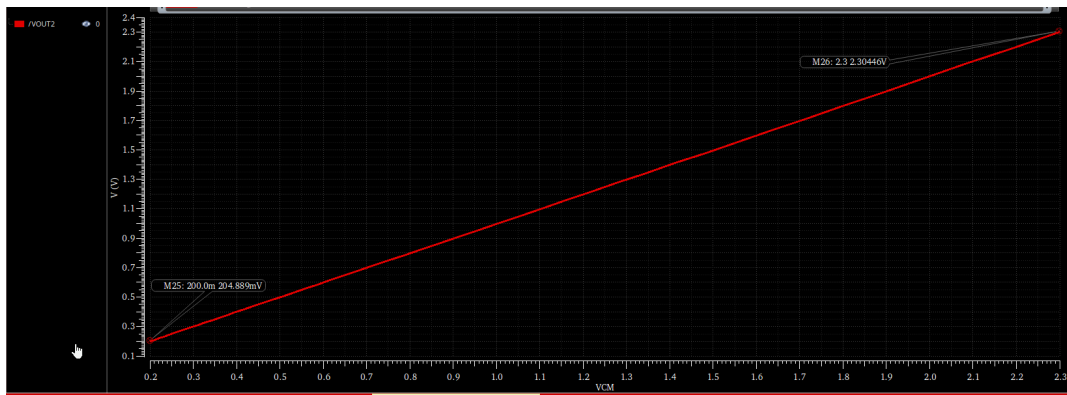


Figure 19: V_O vs. V_{IN} , $R_L = 2\text{ k}\Omega$: 204.889 mV at $V_{IN} = 0.2\text{ V}$, 2.30446 V at $V_{IN} = 2.3\text{ V}$ – essentially unchanged from the $10\text{ k}\Omega$ case.

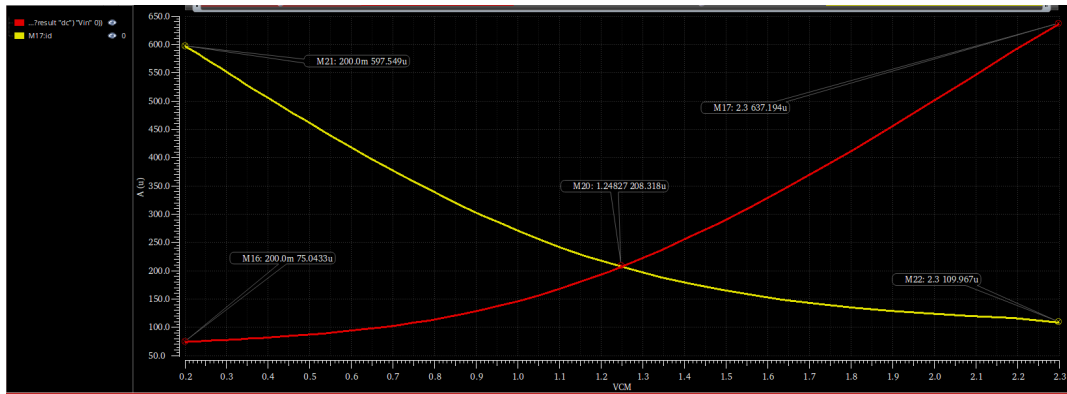


Figure 20: I_{D25} and I_{D26} vs. V_{IN} , $R_L = 2\text{ k}\Omega$, crossing near $V_{IN} \approx 1.248\text{ V}$, $I_D \approx 208.3\text{ }\mu\text{A}$ – same quiescent point as the $10\text{ k}\Omega$ case.

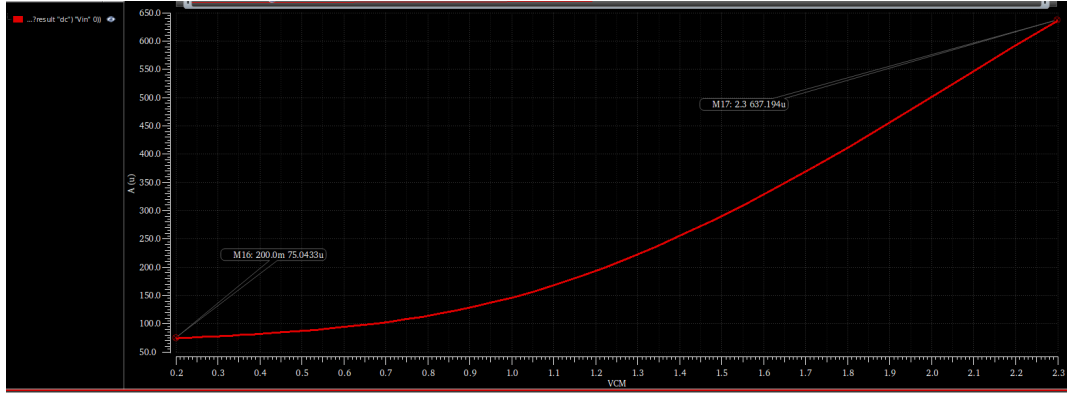


Figure 21: I_{D25} alone: $75.043 \mu\text{A}$ at $V_{IN} = 0.2 \text{ V}$ rising to $637.194 \mu\text{A}$ at $V_{IN} = 2.3 \text{ V}$.

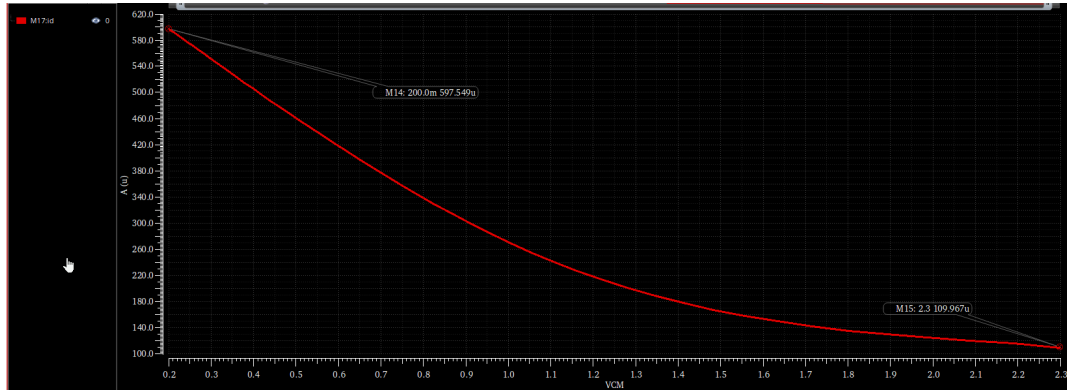


Figure 22: I_{D26} alone: $597.549 \mu\text{A}$ at $V_{IN} = 0.2 \text{ V}$ falling to $109.967 \mu\text{A}$ at $V_{IN} = 2.3 \text{ V}$.

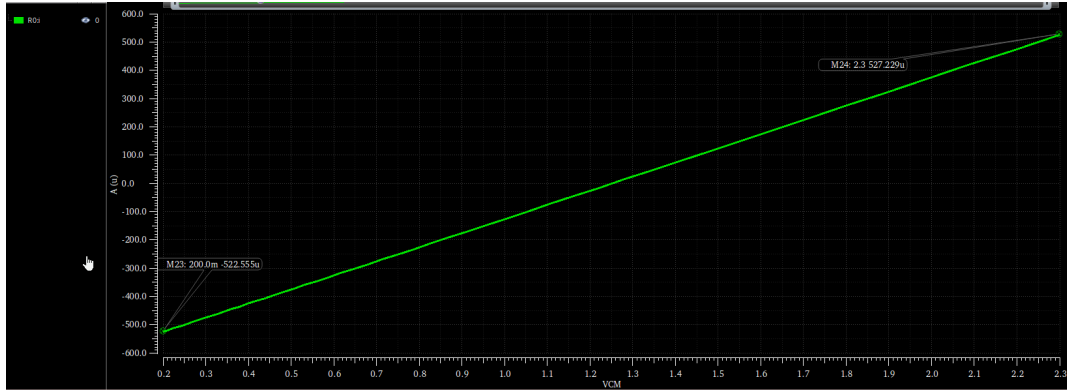


Figure 23: I_{RL} vs. V_{IN} , $R_L = 2 \text{ k}\Omega$: $-522.555 \mu\text{A}$ at $V_{IN} = 0.2 \text{ V}$, $+527.229 \mu\text{A}$ at $V_{IN} = 2.3 \text{ V}$.

Table 3: DC sweep summary, TB1, $V_{IN} : 0.2 \rightarrow 2.3 \text{ V}$

Quantity	Hand Estimate	$R_L = 10 \text{ k}\Omega$	$R_L = 2 \text{ k}\Omega$
V_O @ $V_{IN} = 0.2 \text{ V}$	$\approx 0.2 \text{ V}$	204.6 mV	204.9 mV
V_O @ $V_{IN} = 2.3 \text{ V}$	$\approx 2.3 \text{ V}$	2.3046 V	2.3045 V
Quiescent $I_{D25,26}$ @ V_{MID}	$\approx 200 \mu\text{A}$	207.9 μA	208.3 μA
I_{RL} @ $V_{IN} = 0.2 \text{ V}$	$-105 \mu\text{A}$	$-104.5 \mu\text{A}$	$-522.6 \mu\text{A}$
I_{RL} @ $V_{IN} = 2.3 \text{ V}$	$+105 \mu\text{A}$	$+105.5 \mu\text{A}$	$+527.2 \mu\text{A}$

6.2.3 Stability vs. Common-Mode Voltage

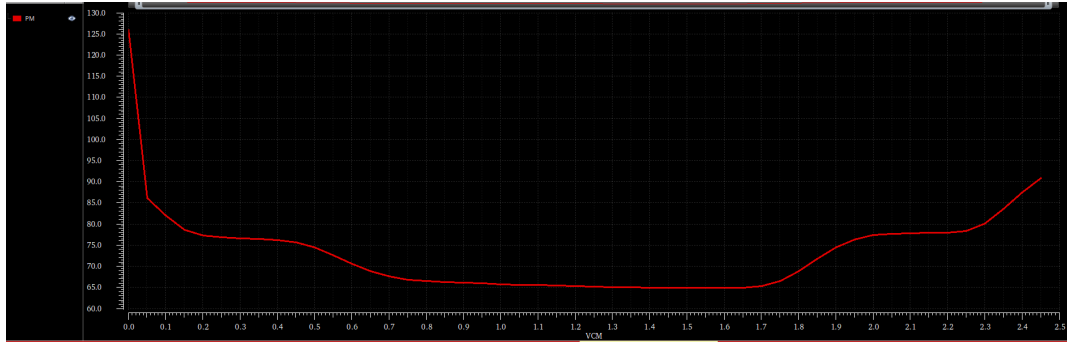


Figure 24: Phase margin vs. V_{ICM} , swept 0 to 2.5 V. PM stays around 64–65° through the central common-mode range and only rises sharply near the rails (edge artifacts / input pair hand-off region).

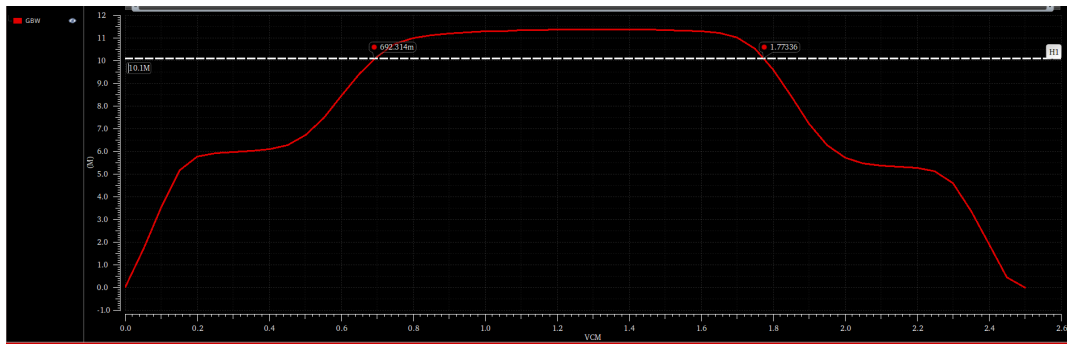


Figure 25: GBW vs. V_{ICM} . GBW stays above the 10.1 MHz reference line for V_{ICM} between 0.692 V and 1.773 V, plateauing around 11 MHz in the middle of that range.

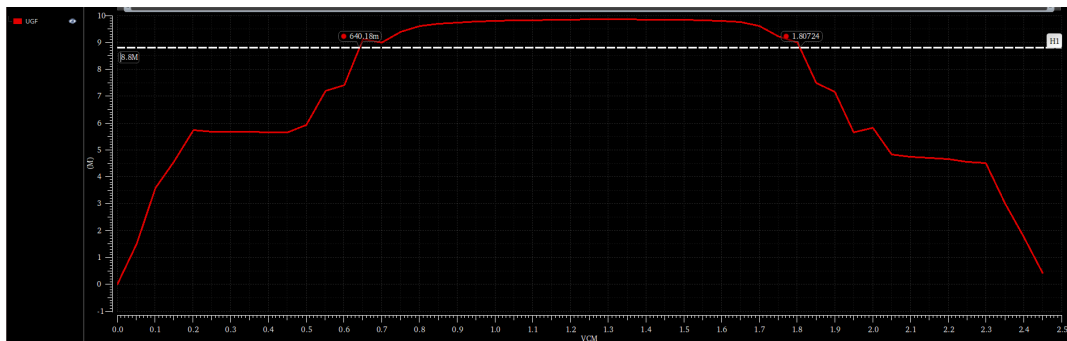


Figure 26: Unity-gain frequency vs. V_{ICM} . UGF stays above the 8.8 MHz reference line for V_{ICM} between 0.640 V and 1.807 V.

7 Transient Analysis

7.1 TB1: Step Response and Slew Rate

7.1.1 Simulation Results

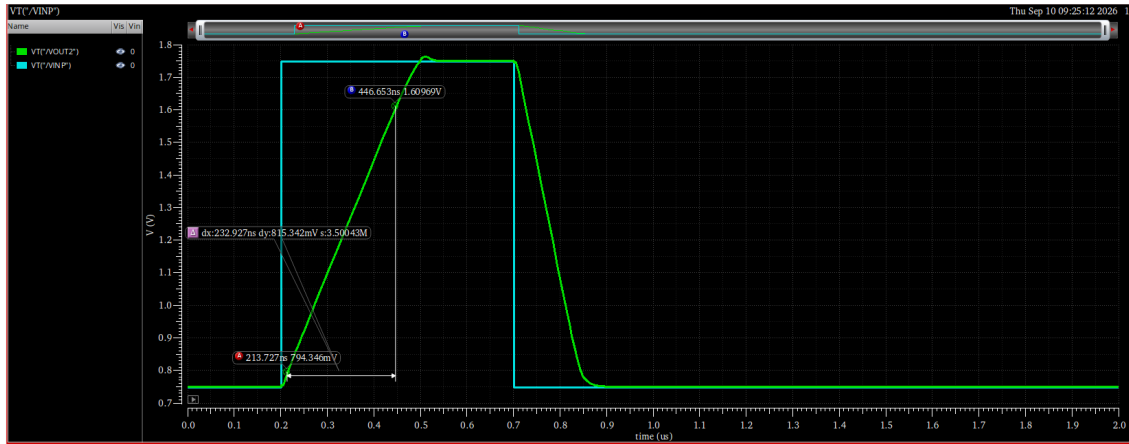


Figure 27: Positive step response in TB1 ($0.75 \text{ V} \rightarrow 1.75 \text{ V}$). The measured positive slew rate (SR^+) is approximately $3.50 \text{ V}/\mu\text{s}$.

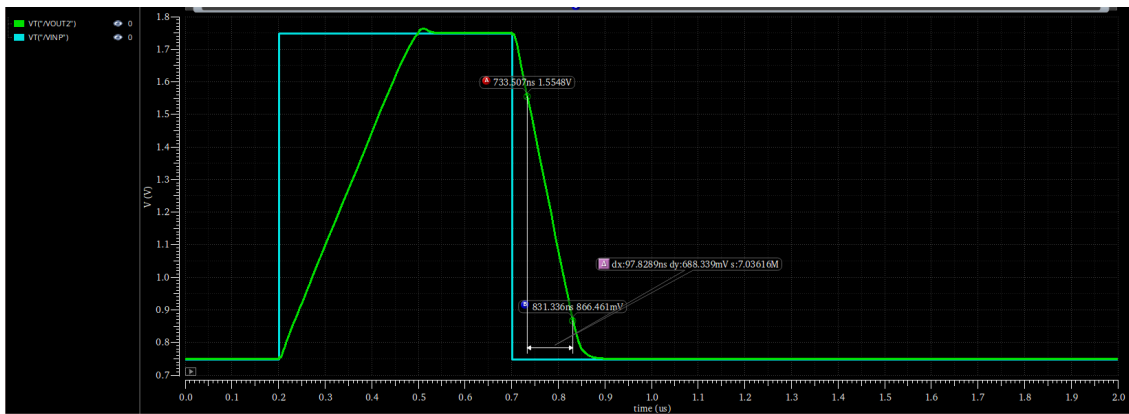


Figure 28: Negative step response in TB1 ($1.75 \text{ V} \rightarrow 0.75 \text{ V}$). The measured negative slew rate (SR^-) is significantly faster at approximately $7.04 \text{ V}/\mu\text{s}$, showcasing the asymmetric dynamic drive capability of the class-AB stage.

7.2 TB2 and TB3: Large Signal Sine Response

7.2.1 Simulation Results

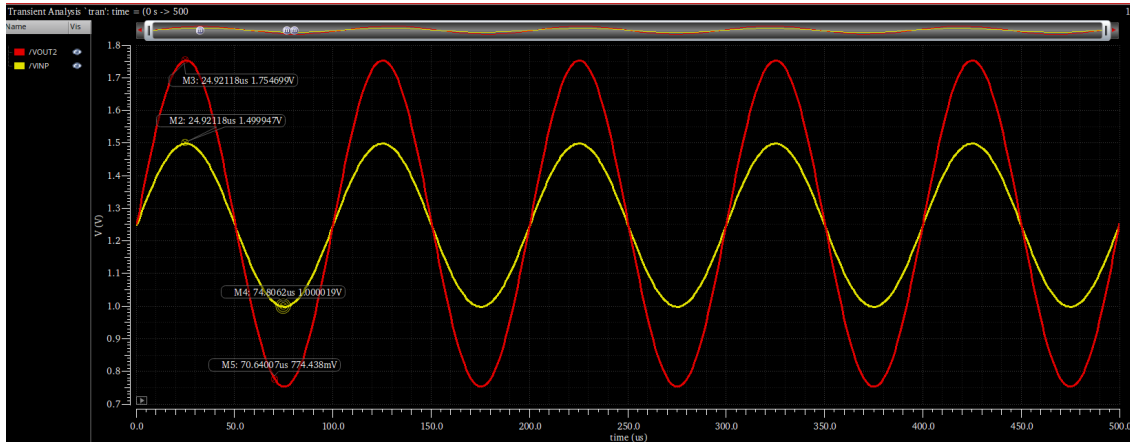


Figure 29: Transient response for TB2 ($G = +2$) using a 250 mV peak input sine wave at 10 kHz. The output swing (red) cleanly doubles the input swing (yellow), operating between 0.75 V and 1.75 V with no visible distortion.

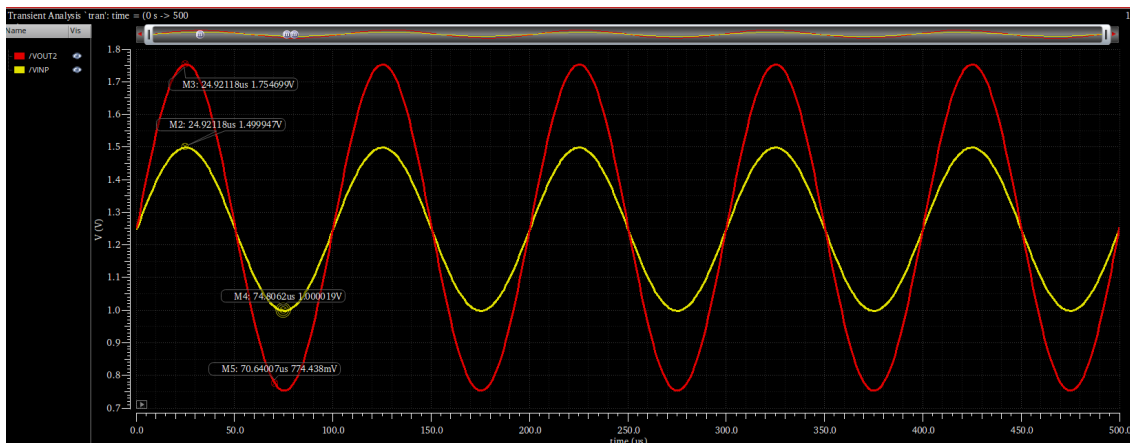


Figure 30: Transient response for TB2 ($G = +2$) using a 250 mV peak input sine wave at 10 kHz. The output swing (red) cleanly doubles the input swing (yellow), operating between 0.75 V and 1.75 V with no visible distortion.

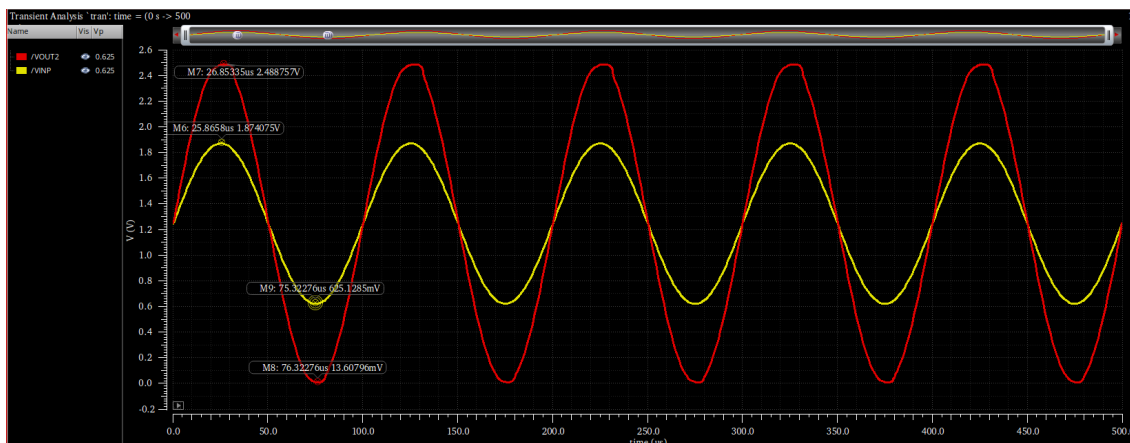


Figure 31: Maximum output swing test for TB2

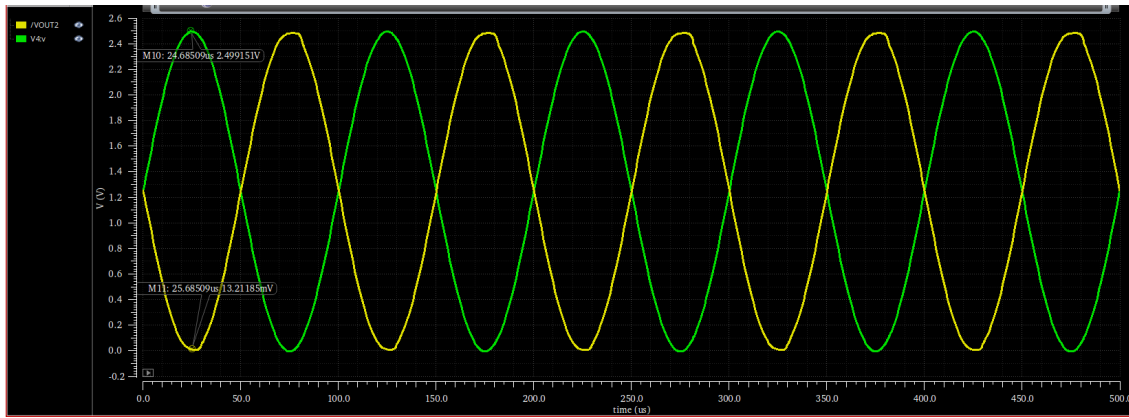


Figure 32: Maximum output swing test for TB3 ($G = -1$). Pushing the input amplitude reveals the robust rail-to-rail capability of the class-AB output stage, with the output successfully swinging from near 0 V up to nearly 2.5 V before clipping.